Yuyuan Han

Computational imaging & robust image-free sensing

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PhD student working on computational imaging and robust image-free sensing, from optical measurement design and reconstruction to networks that infer directly from measurements.

Education & Research Appointments

Xi'an Jiaotong University · PhD student — 2022–present
Electronic Science and Technology · Master's-to-PhD track · Advisor: Huaibin Zheng

Politecnico di Milano · Visiting PhD student — Oct 2026–present
Department of Physics · Supervisor: Antonio Pifferi
Research focus: time-domain diffuse optical tomography for 3D brain imaging

Northwest University · B.Eng. — 2018–2022
Electronic Information Science and Technology

Selected Research

The Lift Spectrum · Robust image-free sensing — Preprint

arXiv · 2026 · Paper · Code

- Image-free single-pixel segmentation can outperform reconstruction-first inference under measurement noise with clean-trained models; introduced a content-adaptive 1D-to-2D lift and task-prioritized loss scheduling.
- 3.1–21.7 percentage-point foreground mIoU gains at 3.13% sampling and 20 dB Gaussian measurement noise across three simulated datasets; optical-bench proof of concept.

Sparse Hadamard single-pixel imaging

Optics Express · 2025 · Paper · Code

- Tunable sparse Hadamard patterns retain orthogonality and improve noise resilience, with two training-free modes and a sparsity-dependent noise model tested in simulations and through-scattering optical experiments.

LDGI · Ghost imaging for optical phased-array line patterns

Optics Communications · 2025 · Paper · Code

- Loop differential ghost imaging reduces spatial noise asymmetry for 1D optical phased-array line modulation; validated with a noise-symmetry metric, numerical simulations and spatial-light-modulator-generated line patterns.

Current Research

Time-domain diffuse optical tomography for 3D brain imaging; acquisition-aware single-pixel sensing on an optical bench; exploratory question answering from measurements with a frozen vision–language model.

Open Source & Tools

Open source · remote-gpu-trainer, an agent skill for Claude Code and Codex that runs, verifies and delivers GPU experiments
Tools & lab · PyTorch, MATLAB, COMSOL; DMD/SLM illumination, single-pixel detection, optical alignment and system integration
Device modeling · Geometry optimization of electro-optic phased arrays for beam steering

Award & Service

2024 · Second Prize, National Post-Graduate Mathematical Contest in Modeling · Code
Reviewer · IEEE TIP, IEEE TCI, Optics Express, Optics Letters (2025–2026)